Data Visualisation in R: Graphs

In this tenth article in the R series, we will continue to explore data visualisation in R with the lattice and ggplot2 packages.

We will be using the R version 4.1.2 installed on Parabola GNU/Linux-libre (x86-64) for the example code snippets in this article.

```
$ R --version
R version 4.1.2 (2021-11-01) -- "Bird Hippie"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

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Lattice

Line chart

Consider the consumer prices (annual per cent) inflation data for India between 1960 and 2022 available from the World Bank. You can use the years in the x-axis, and the inflation on the y-axis to produce a line chart using the *xyplot* function, as shown below:

```
> x<-c(1960:2020)

> y<-c(1.77,1.69,3.63,2.94,13.35,9.47,10.80,13.06,3.23,-0.58,5.09,3.07,6.44,16.94,28.59,5.74,

-7.63,8.30,2.52,6.27,11.34,13.11,7.89,11.86,8.31,5.55,8.72,8.80,9.38,7.07,8.97,13.87,11.78,6.32,10.24,10.22,8.97,7.16,13.23,4.66,4.00,3.77,4.29,3.80,3.76,4.24,5.79,6.37,8.34,10.88,11.98,8.85,9.31,11.06,6.64,4.90,4.94,3.32,3.94,3.72,6.62)

> d <- data.frame(x,y)

> xyplot(y~x, data=d, type="l", main="Inflation, consumer prices (annual %)")
```

The line chart is shown in Figure 1.

The *xyplot* accepts the following arguments: